

Data Analysis I: Introduction to Data Science with R

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CM	TD	TP	TPE	Credit hours	Credits
24	9	8	4	45	3

Course Goals and Objectives

“The goal is where we want to be. The objectives are the steps needed to get there.”

Goals

This course is designed to give students the necessary skills to analyze data from research projects.

This course provides the participants with a practical application of R

Students will review several statistical techniques, gain an understanding of when and why to use these various techniques as well as how to apply them with confidence, interpret their output, and graphically display the results.

Course Outline

Unit I: An overview of data science

Unit II: Scales of Measurement

Unit III: Introduction to R

Unit IV: Data analysis in R

- Descriptive analytics with R

Mean, Mode, Median, Standard deviation, ...

- Exploratory Analytics with R

- Predictive Analytics with R

✓ Types of Learning:

- Linear Regression- Simple Linear Regression
- Implementation in R - functions on `lm()` - `predict()` - plotting and fitting regression line.
- Multiple Linear Regression - Introduction -comparison with simple linear regression
- Principal Component Analysis

Unit V: ANOVA and applications in R

Course Requirements

- Access to a computer with an internet connection
- Ability and permissions to download files & install software
- Basic knowledge of English (Course is delivered in English language)

ASSESSMENT COMPONENTS FOR COURSE GRADE

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- Attendance + Assignments: 2/3 of grade.
- Effective class participation = 1/3 of grade
- You are considered absent should you be 30 minutes late
- Three absences, you lose 30% of the final CA mark
- Final Written Exam= 70% of grade!!!!
- Please respect deadlines!!!

Lecture I

An overview of data science

Motivating Questions

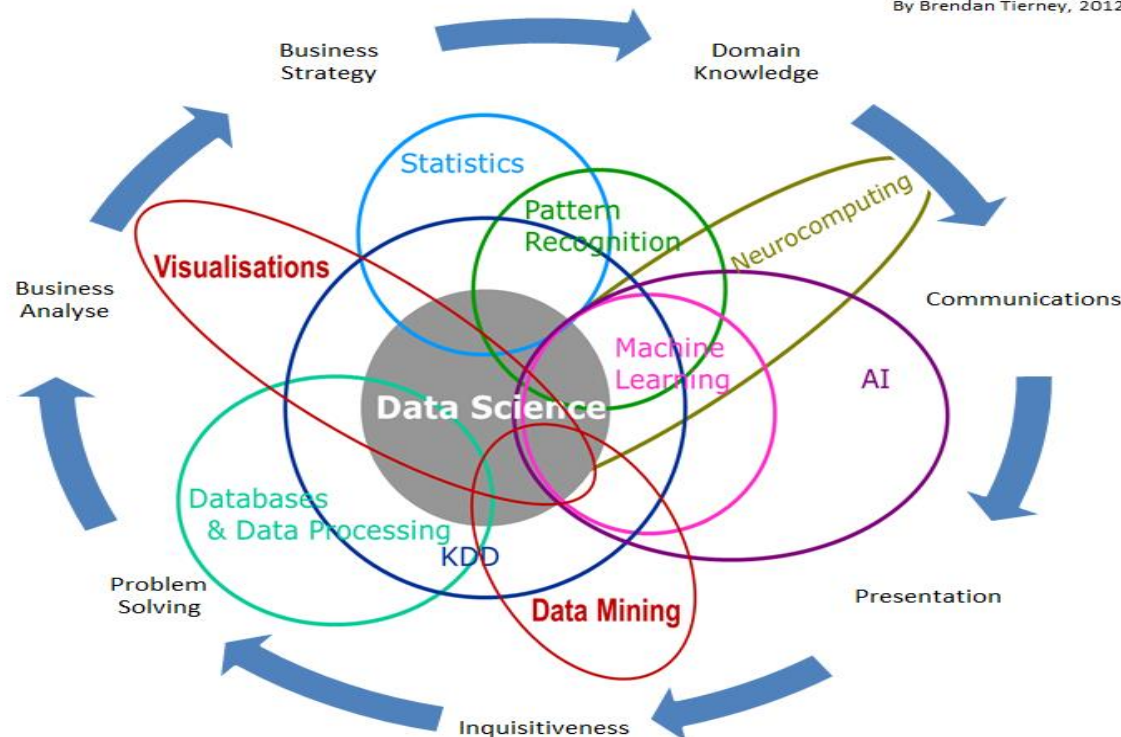
- What is Data Science ?
- Who is Data Scientist?
- What Data Scientist do?
- Data Scientist Type?
- Why we need them?

What is Data Science (DS)?

- An area that *manages, manipulates, extracts, and interprets* knowledge from tremendous amount of data
- A multidisciplinary field of study that combines programming skills, domain expertise and knowledge of statistics and mathematics to extract useful insights and knowledge from data”.
- The goal is to address the challenges in **big data**

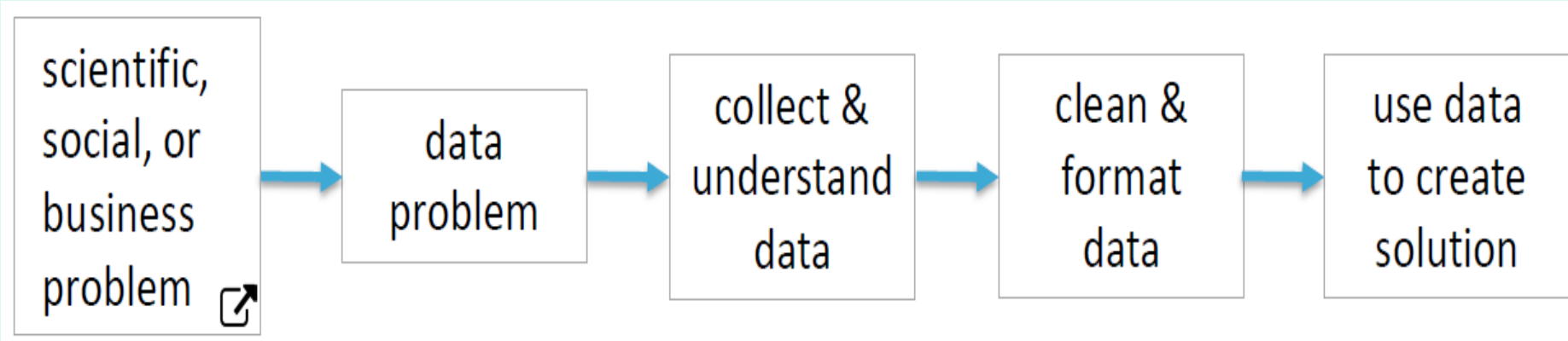
Data Science Is Multidisciplinary

By Brendan Tierney, 2012

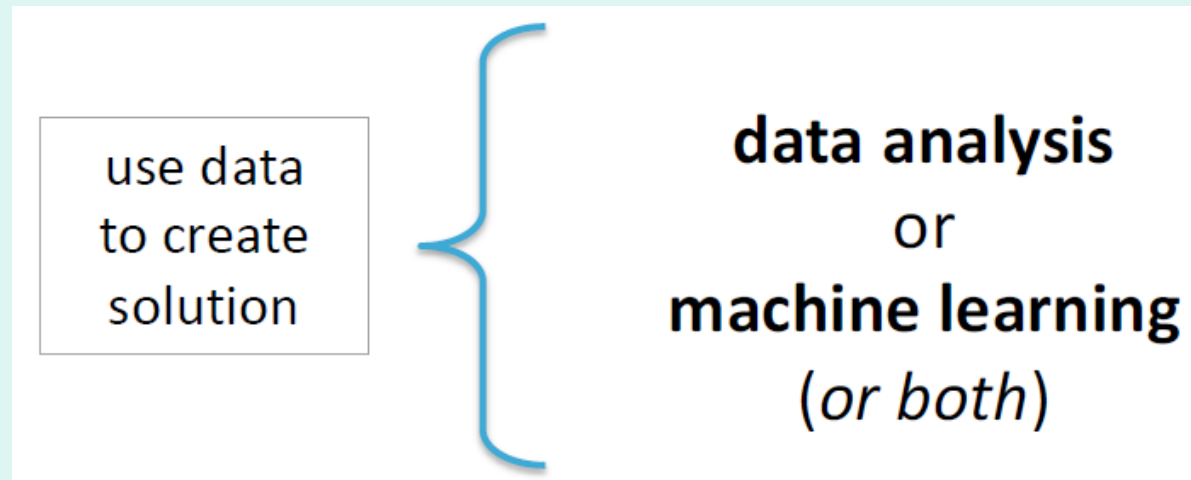


...Data Science ?

...solving problems with data...



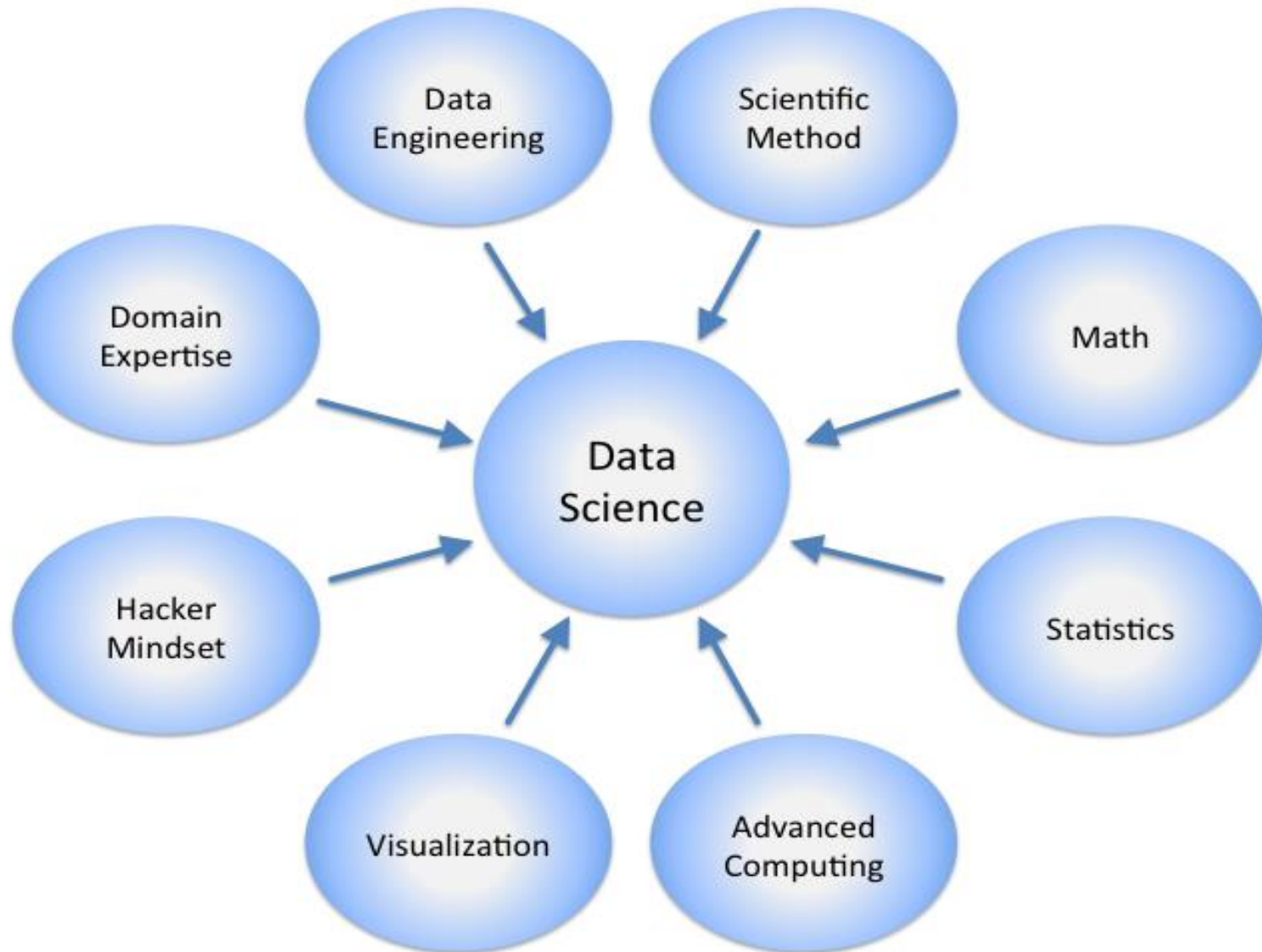
...which step is most challenging?



Concentration in Data Science

- Mathematics and Applied Mathematics
- Applied Statistics/Data Analysis
- Solid Programming Skills (R, Python, Julia, AWS, SQL)
- Data Mining
- Data Base Storage and Management
- Machine Learning and discovery

...Data Science

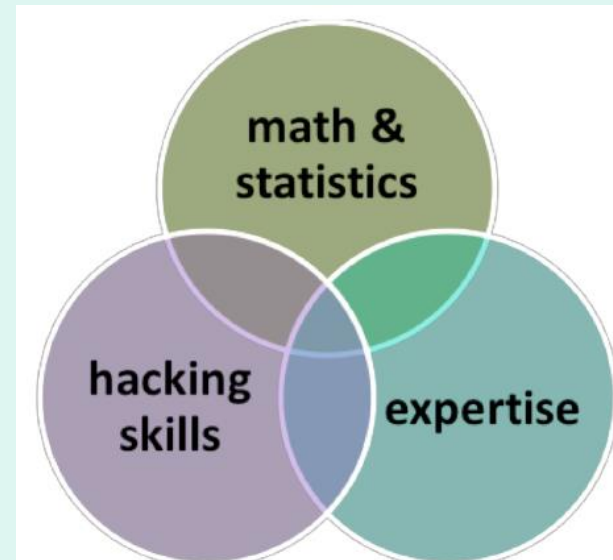


Data Scientists

Data Scientist (n.): Person who is better at statistics than any software engineer and better at software engineering than any statistician.

They combine a wide range of skills and modern technologies to analyze data collected from sensors, customers, smartphones, the web, and other sources.

What makes a good data scientist?



...Data Scientists

- Data Scientist
 - The Sexiest Job of the 21st Century
- They find stories, extract knowledge. They are not reporters
- Data scientists are the key to realizing the opportunities presented by big data. They bring structure to it, find compelling patterns in it, and advise executives on the implications for products, processes, and decisions



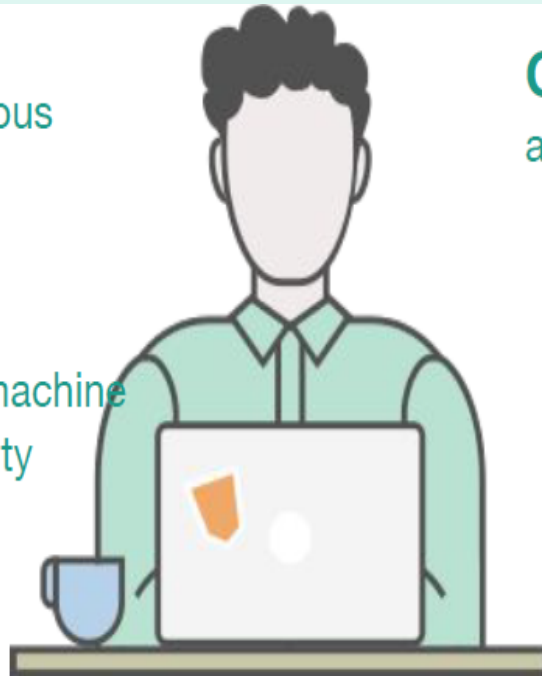
What do Data Scientists do?

- National Security
- Cyber Security
- Business Analytics
- Engineering
- Healthcare
- And more

What does a good data scientist look like?

Inquistive – skeptical and curious

Knowledgeable – knows machine learning, statistics, and probability



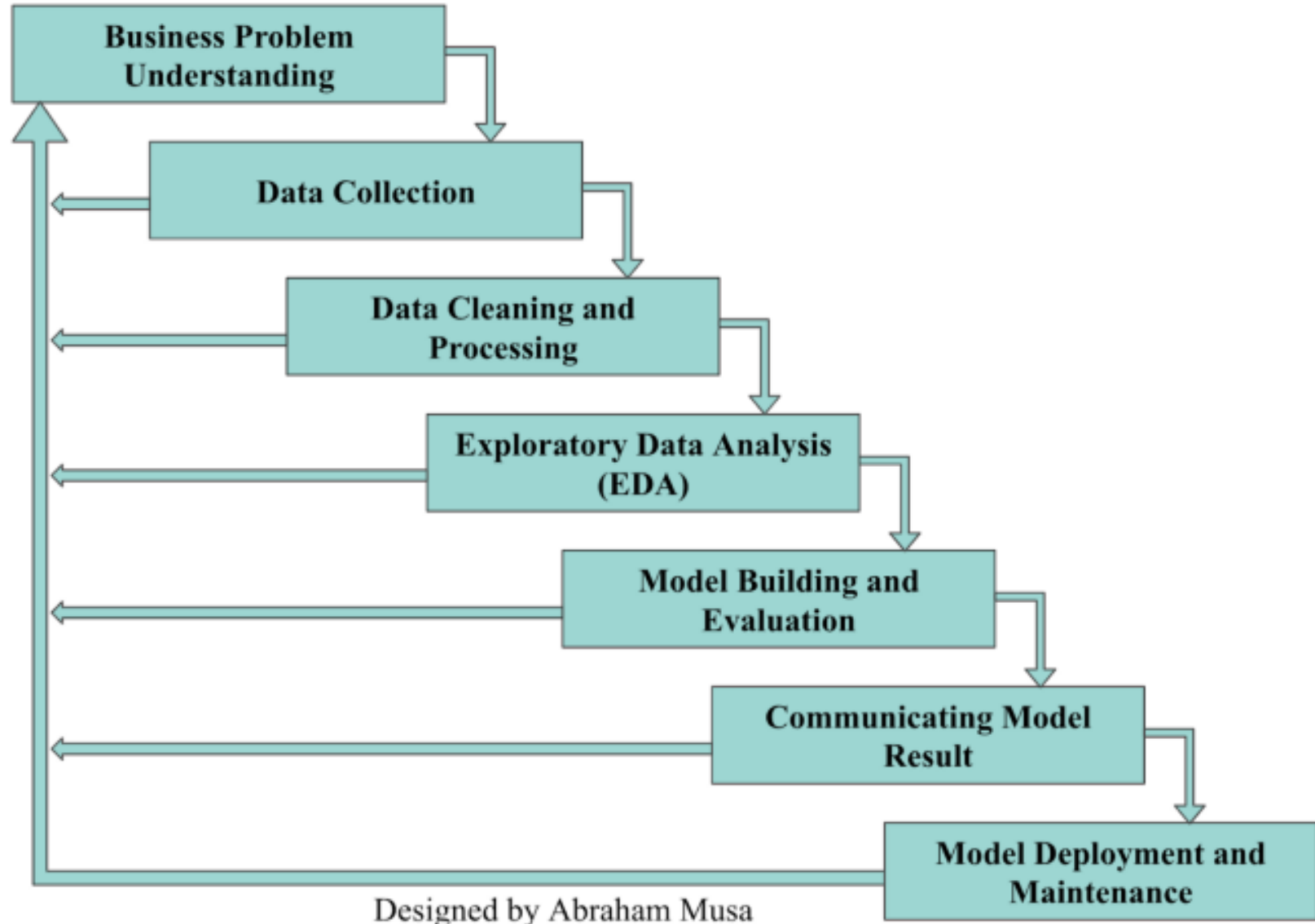
Coding – is good at coding, hacking, and general programming

Product Oriented – knows how to build data products and visualizations to make data understandable to mere mortals

Scientific Method – Creates hypotheses, tests them, and updates understanding

Domain Knowledge – understands the business and how to tell the relevant story from business data. Able to find answers to *known unknowns*.

Data Science Project Life Cycle



What is data analysis?

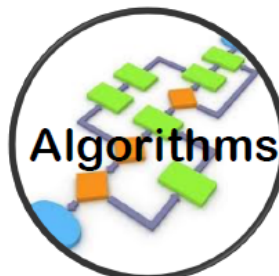
...using data to discover useful information...



- **data:** anything you can *measure* or *record*



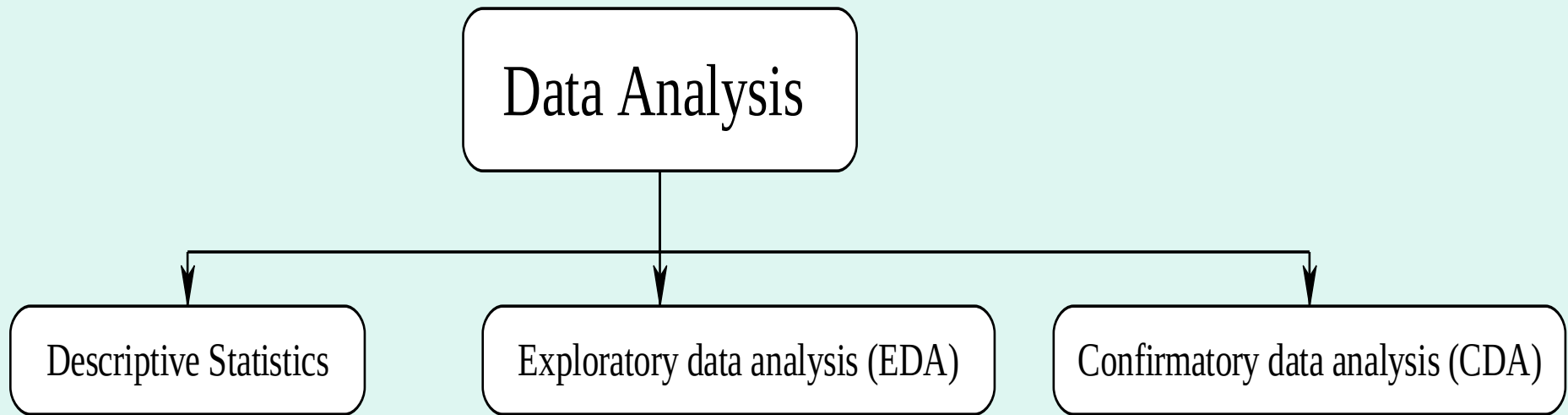
- **statistics:** summarize (and visualize) *main characteristics* of the data



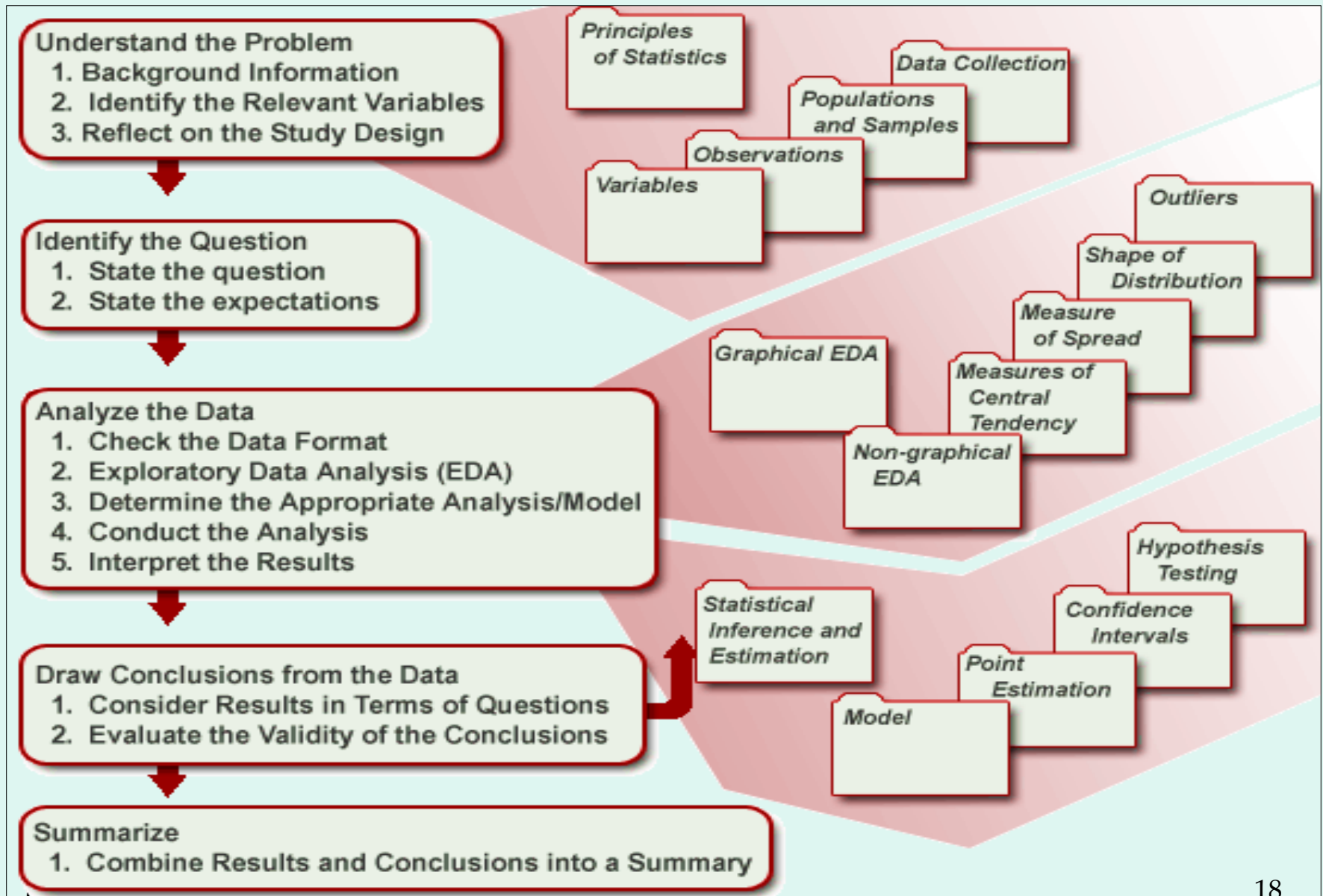
- **algorithms:** apply algorithms to find *patterns* in the data

Activate W
Go to Settings

Divisions of Data Analysis



General approach to data analysis



Key requirements for becoming a Data Analyst:

1. Be well-versed with programming languages (XML, Javascript, or ETL frameworks), databases (SQL, SQLite, Db2, etc.), and also have extensive knowledge on reporting packages (Business Objects).
2. Be able to analyze, organize, collect and disseminate *Big Data* efficiently.
3. You must have substantial technical knowledge in fields like database design, data mining, and segmentation techniques.
4. Have a sound knowledge of statistical packages for analyzing massive datasets such as R, Python, SAS, Excel, and SPSS, ...

What is machine learning?

...creating and using models that learn from data...



- **data:** anything you can *measure* or *record*



- **model:** specification of a (mathematical) *relationship* between different variables



- **evaluation:** how well does the model *work?*

Examples

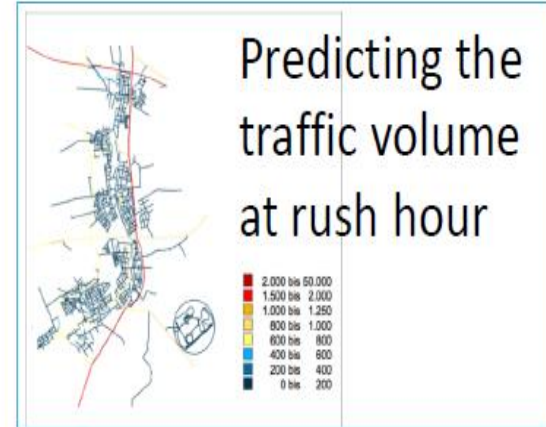
Identifying zip code
from handwritten
digits



Detecting
communities
in social
networks



Predicting the
traffic volume
at rush hour



Detecting fraudulent
credit card
transactions

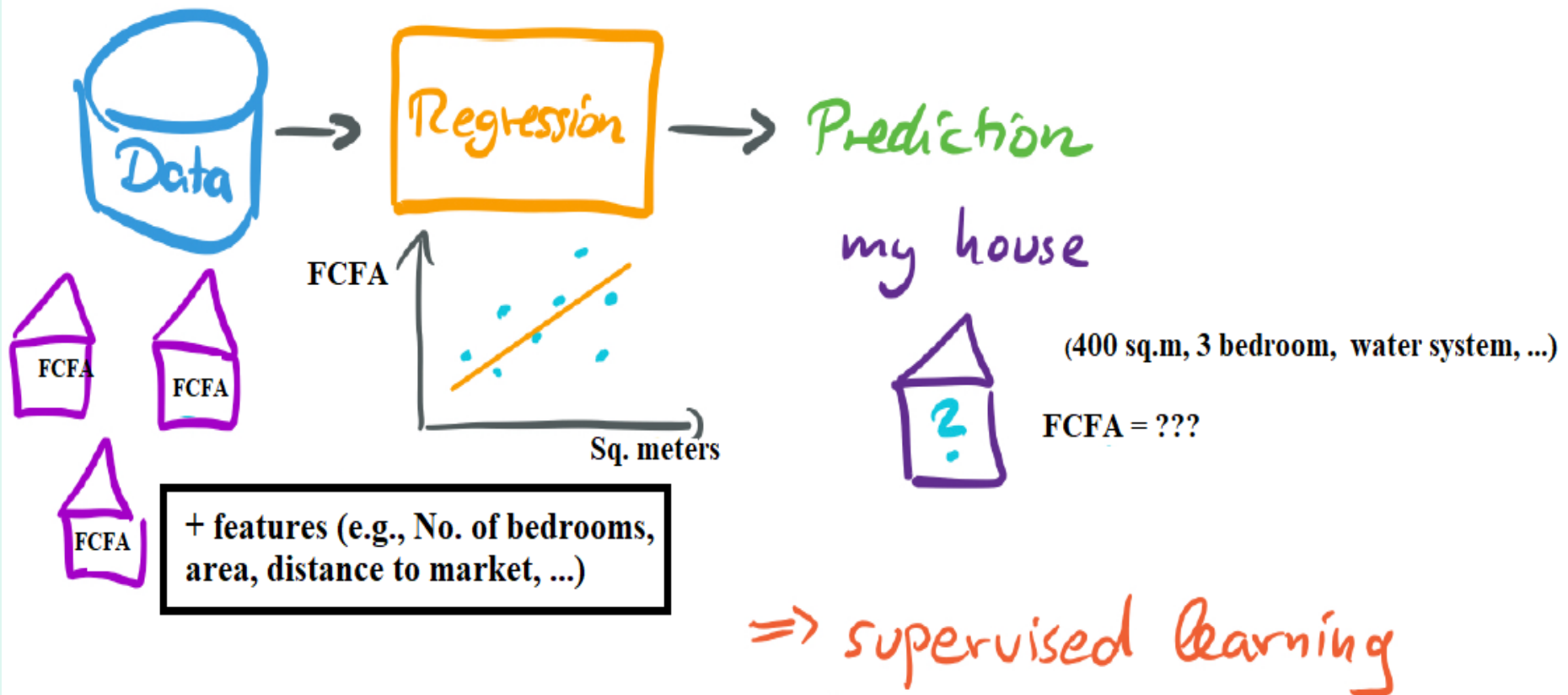


Determining the
location of distribution
centers based on
customers'
residence



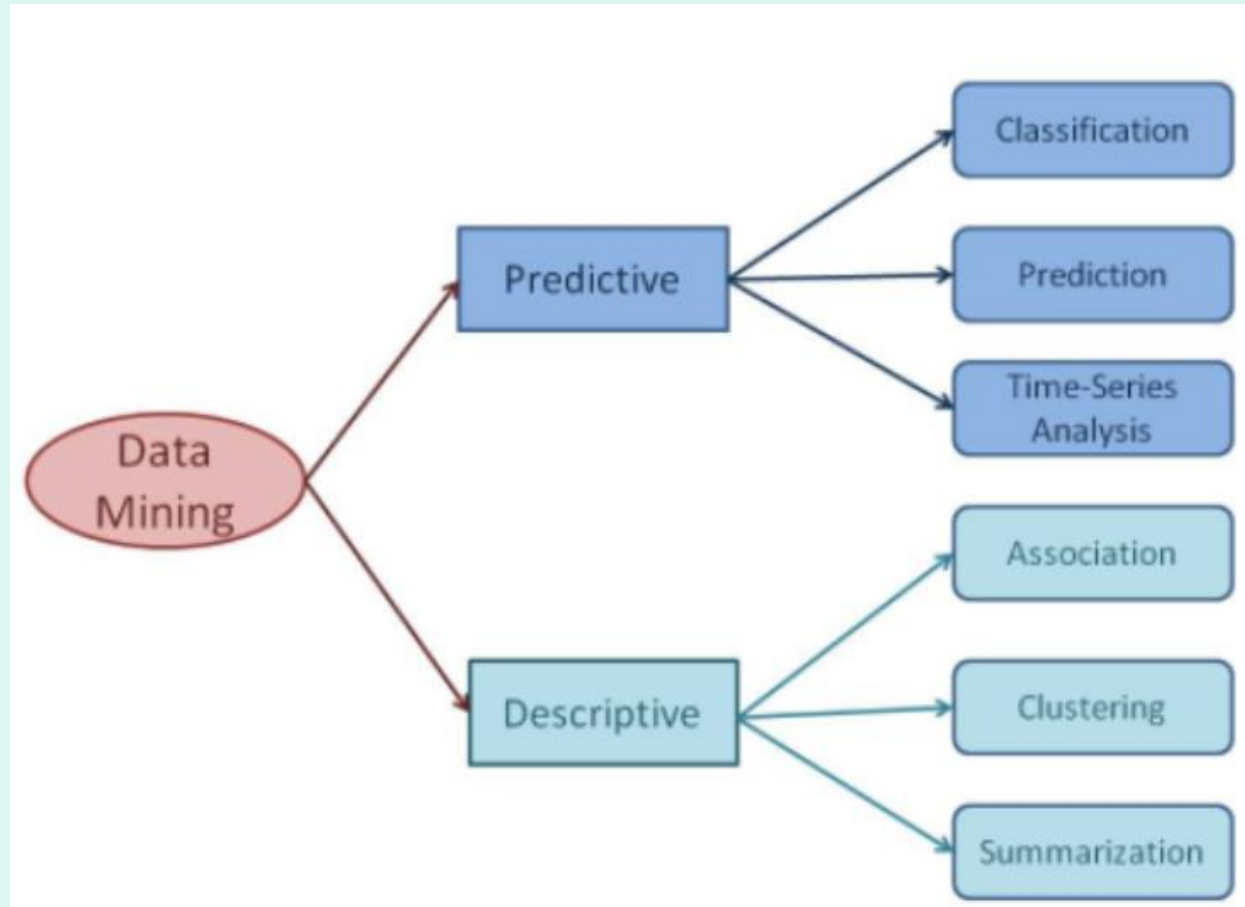
Learning from data

a) Regression



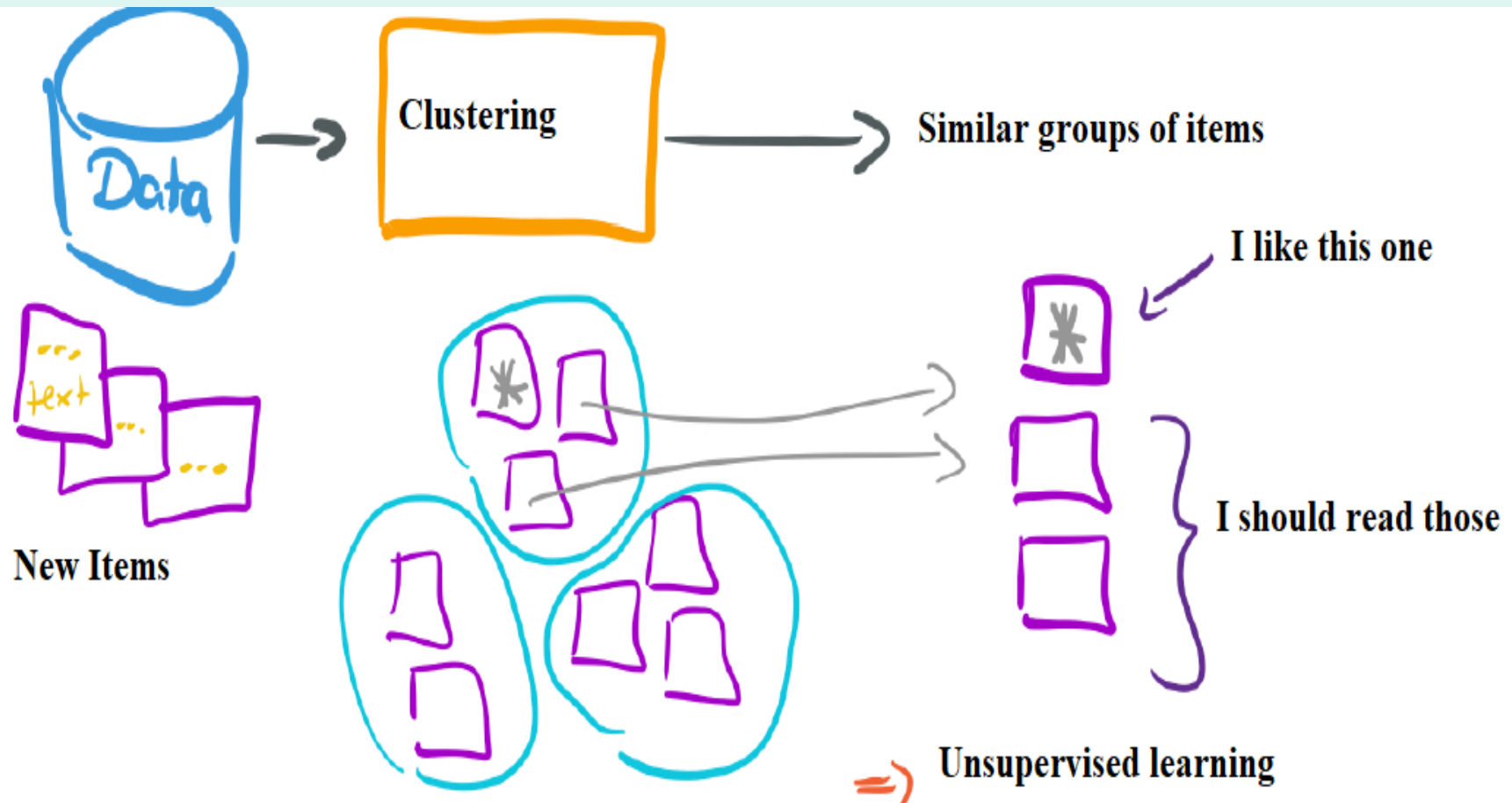
b) Classification in Data Mining

⇒ supervised learning



input data is provided to the model along with the output

c) Clustering



Unsupervised learning refers to the use of artificial intelligence (AI) algorithms to identify patterns in data sets containing data points that are neither classified nor labeled.

What is machine learning?

regression, classification, clustering

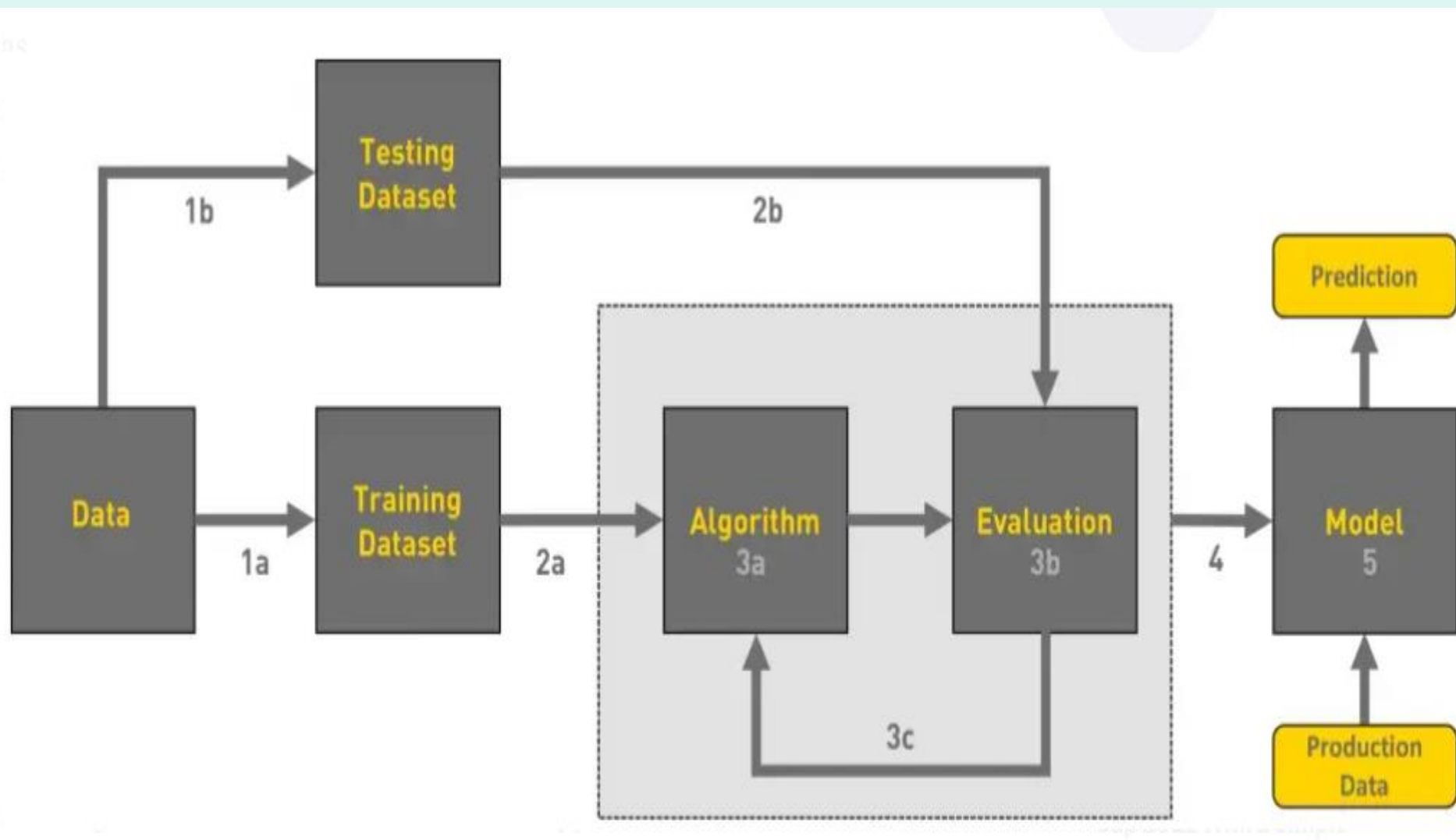
...creating and using models that learn from data

→ supervised learning/predictive modelling

- come up with predictions
- extract knowledge/insights

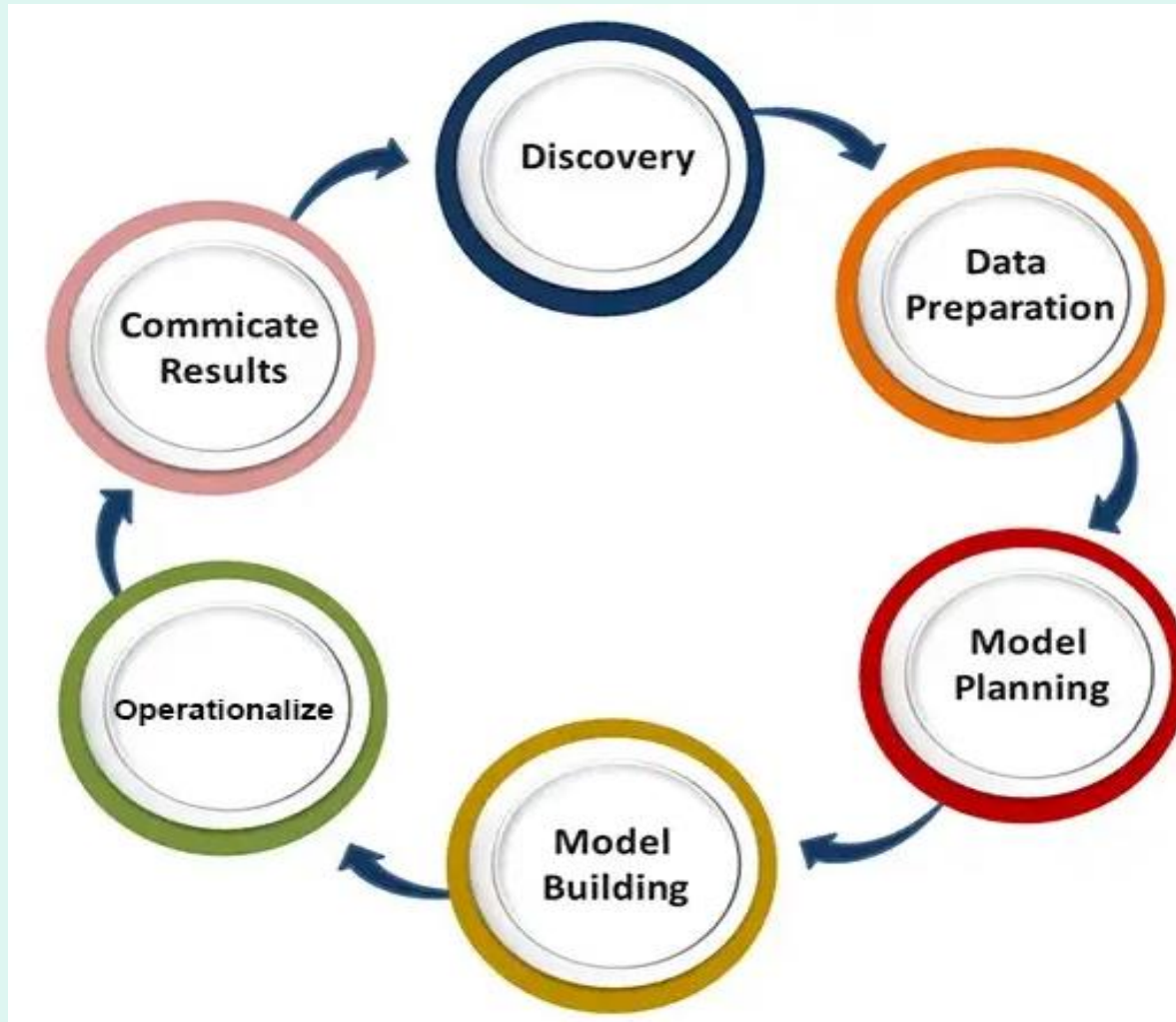
→ unsupervised learning/data mining

Machine learning workflow



Data science lifecycle

A data science lifecycle is defined as **the iterative set of data science steps required to deliver a project or analysis**. There are no one-size-fits that define data science projects. Hence you need to determine the one that best fits your business requirements. Each step in the lifecycle should be performed carefully.



The main phases of data science life cycle

The algorithms and methods that data scientists use to filter data into categories include the following, among others:

Decision trees. ...

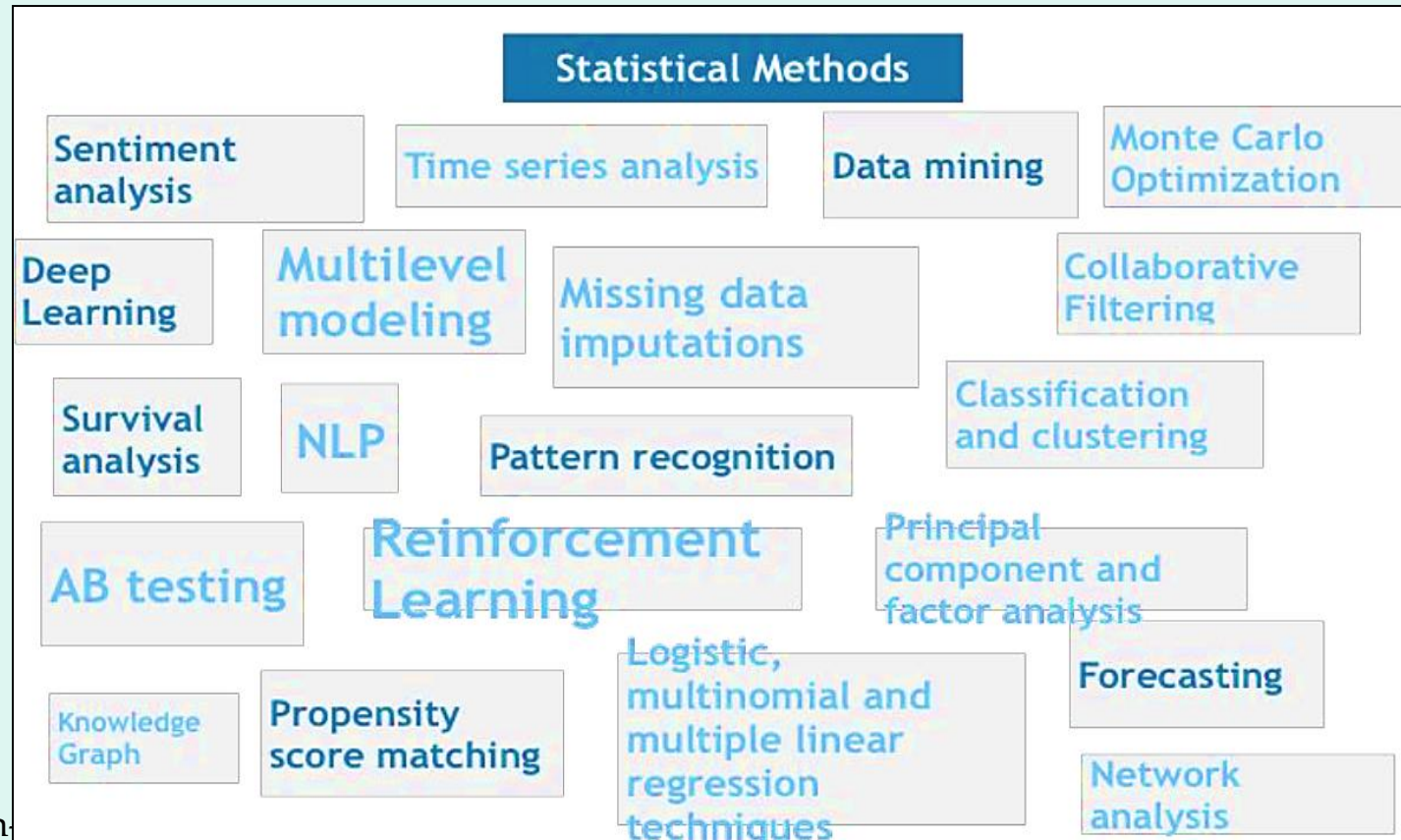
Naïve Bayes classifiers. ...

Support vector machines. ...

K-nearest neighbor. ...

Logistic regression. ...

Neural network



Review Questions

1. What is data science?
2. How is data science different from data analysis?
3. Distinguish between supervised and unsupervised learning algorithms
4. What do you understand by:
 1. Data
 2. Machine learning
 3. Big data
 4. Data mining?

Lecture II

Scales of Measurement

What is Measurement?

- The assignment of numerals to objects or events according to rules.
 - **Numerals** are labels that have no inherent meaning, for example zip codes, or automobile license plates.
 - **Numbers** are numerals that have quantitative meaning and can be analyzed, for example, age.
- ■ The rules for assigning labels to properties of variables are the most important components of measurement, because the result of poor rules is meaningless outcomes.
- ■ Concepts often cannot be measured directly, e.g., “intelligence,” so what is usually measured are indicators of constructs, such as speed, logic, verbal skill, etc.

Levels of Measurement

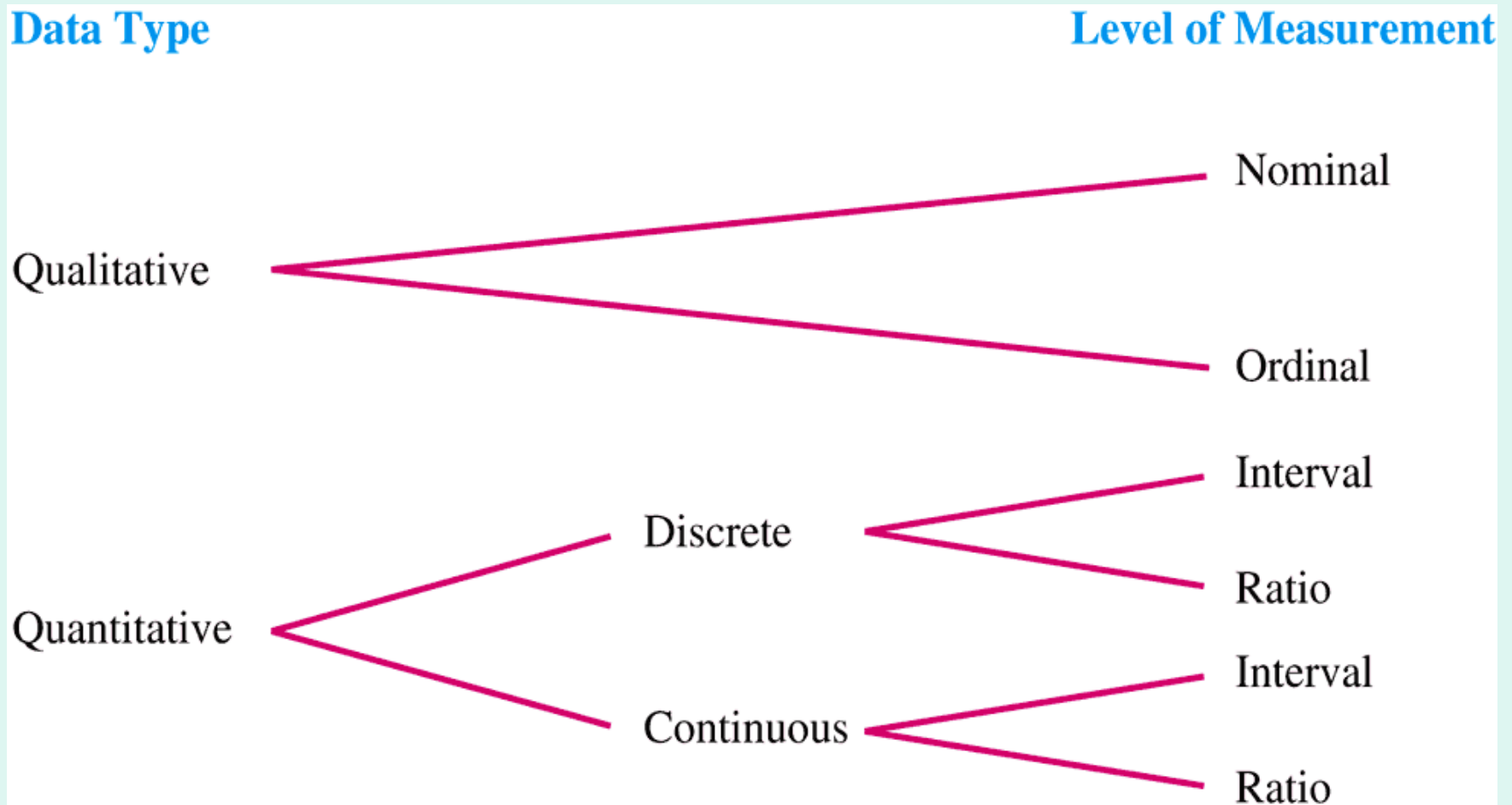
■ Four levels of measurement have been identified:

- **Nominal**
- **Ordinal**
- **Interval**
- **Ratio**

These levels differ in how closely they approach the structure of the number system we use.

- Understanding the level of measurement of variables used in research is important because the level of measurement determines the types of statistical analyses that can be conducted.
- The conclusions that can be drawn from research depend on the statistical analysis used.

Possible data types and levels of measure.



***The type of data you have dictates the type of analysis you will perform.

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Nominal Scale

- In nominal measurement, all observations in one category are alike on some property and differ from the members in the other category on that property (e.g., sex, marital status).
- Ordering of categories exists. We cannot say one category is better or worse, or more or less than another.

Nominal — Numbers used as Names

- Basic Empirical Operations
 - Determination of equality
- Permissible Statistics
 - Number of cases
 - Mode
 - Contingency correlation
- Examples
 - Numbers on football jerseys
 - Assignment of type or model numbers to classes

Ordinal Scales: This scale has the characteristic of the nominal scale in that different numbers mean different things, but also has the characteristic of "greater or lesser". It measures a variable in terms of magnitude, or rank.

Example:

- socioeconomic
 - class
 - grades
 - preferences
-
- Ordinal scales tell us relative order, but give us no information regarding differences between the categories.
 - For example, runners in the 100 meter dash finish 1st, 2nd, 3rd etc. Is the number of seconds between 1st and 2nd place the same as those between 2nd and 3rd place? Certainly not necessarily.

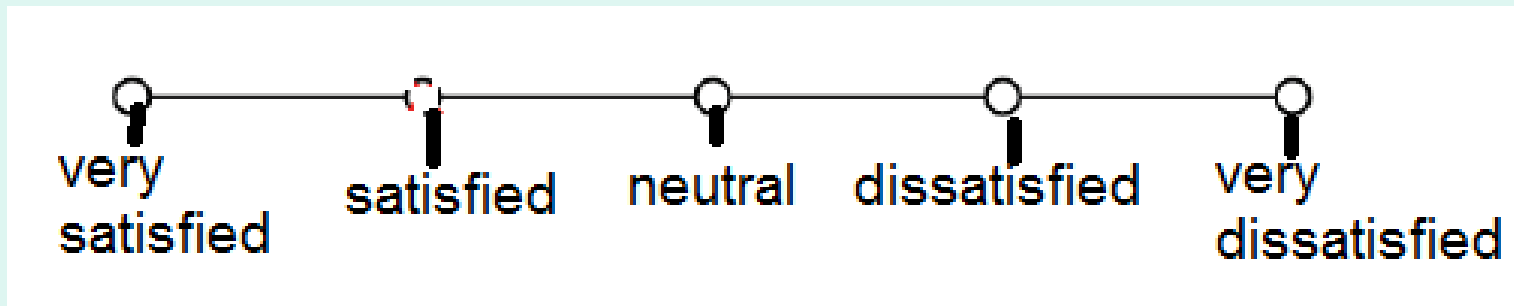
... Ordinal

To convert these features into numerical, integer encoding/ label-encoding can be used. Here, each value will be assigned an integer label such as:

low=0, average=1, medium=2, high=3, very high=4.

... Ordinal Scale: Rank order data

- Most questionnaires use Likert type items. For example, we may ask teachers about their job satisfaction.
- Asking whether a teachers is very satisfied, satisfied, neutral, dissatisfied, or very dissatisfied is using an ordinal scale of measurement.



Interval Scales

- This scale has the properties of the nominal and ordinal scales, but here the magnitude between the consecutive intervals are equal. Temperature is the example that is usually given to illustrate an interval scale.
- When **distance** between attributes has meaning, for example, temperature (in Fahrenheit) - distance from 30-40 is same as distance from 70-80
- * Interval scales do not have a true zero. 0 degrees do not mean the absence of heat (although it might feel like it).

" If a change from 1 to 2 has the same strength as a 4 to 5, then we would call it an interval level measurement **(if not, then it's just an ordinal qualitative measurement)**.

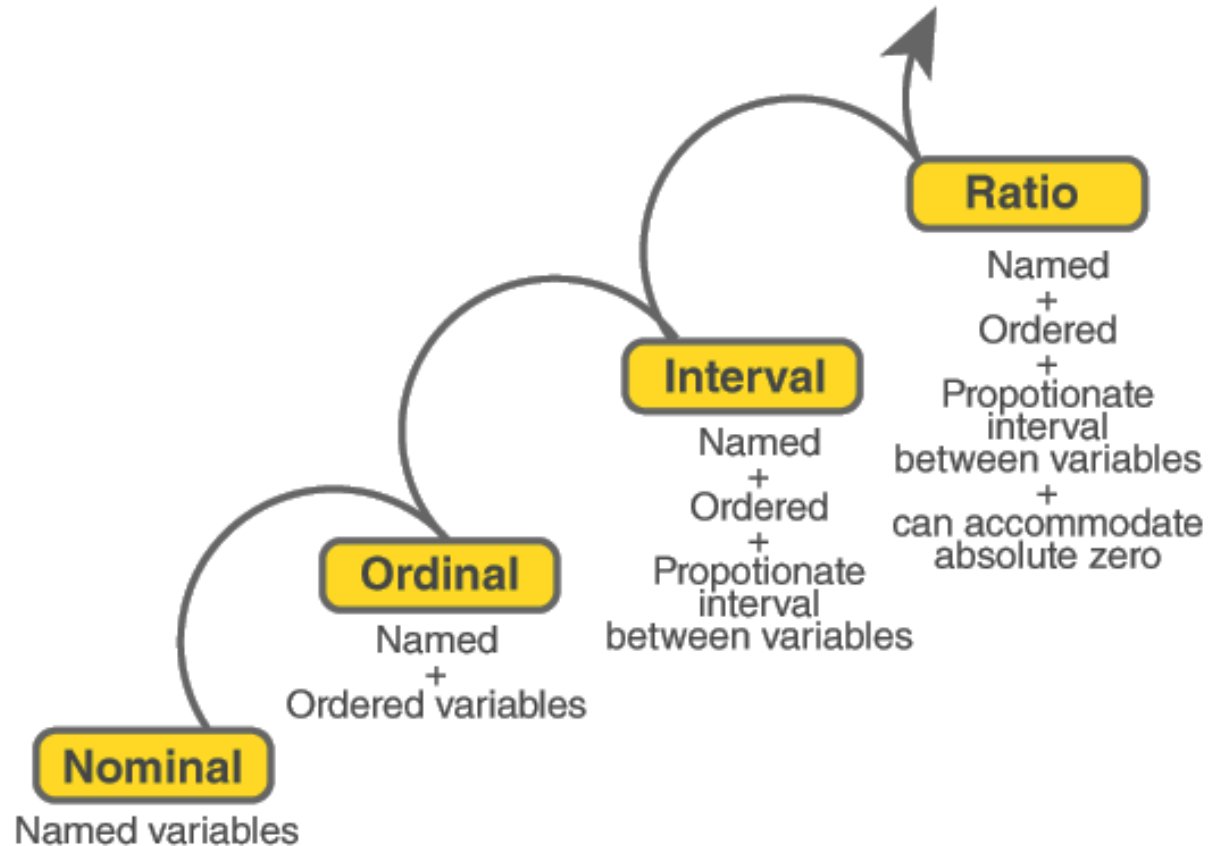
Ratio Scales

- Ratio scales have all of the characteristics of the nominal, ordinal and interval scales. In addition, however, ratio scales have a true zero.
- There are true ratios. One can use all mathematical operations on this scale.
- Examples:

weight
height
time
distance

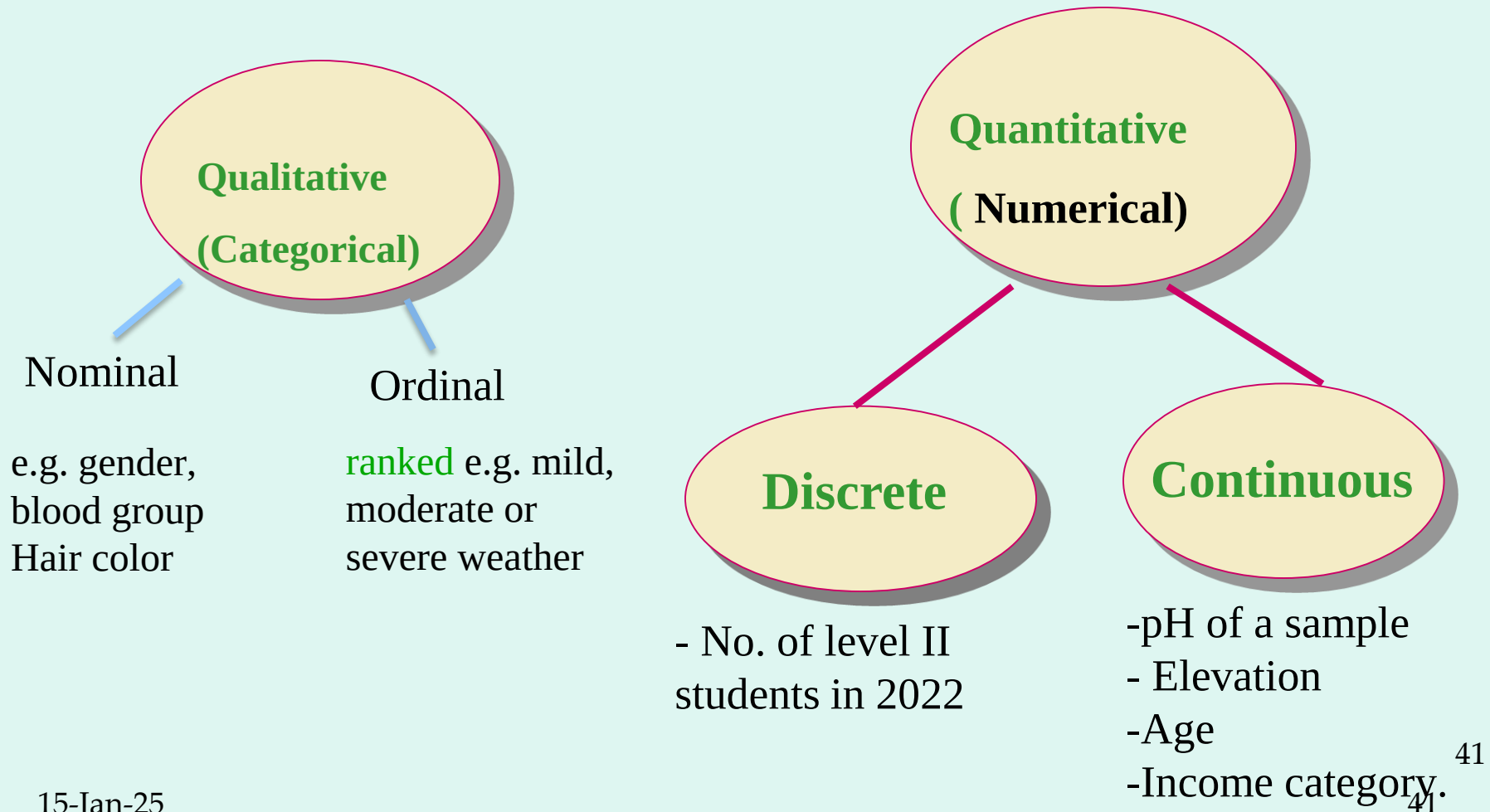
- * 10 miles is twice as long as 5 miles. 0 miles is no distance.
- In our descriptions of data in this course, we will assume that we are using ratio scales most of the time. We call these **PARAMETRIC STATISTICS**.
- However, there will be times when all we have to work with are ordinal scales. When we use these scales, our data will be rank ordered. We will call these **NONPARAMETRIC STATISTICS**.

LEVELS OF MEASUREMENT



Types of Variables

- A **variable** is a characteristic that changes or varies over time and/or for different individuals or objects under consideration.
- **Variables** are the quantities measured in a sample. They may be classified as:



Variables

- Variables can be further classified as:
 - **Dependent/Response**. Variable of primary interest (e.g. blood pressure in an antihypertensive drug trial). Not controlled by the experimenter.
 - **Independent/Predictor**
 - called a **Factor** when controlled by experimenter. It is often nominal (e.g. treatment)
 - **Covariate** when not controlled.
- If the value of a variable cannot be predicted in advance then the variable is referred to as a **random variable**

Level of measurement			
Analysis	Nominal	Ordinal	Interval/ratio
Locate the <i>centre value</i>	Mode, Modal class	Median, Mode, Modal class	Mean, Median, Modal class
Calculate the <i>spread</i>		Range, Inter quartile range	Range, Inter quartile range Standard deviation
Summarize with a <i>table</i> .	Table of frequencies	Table of frequencies	A frequency table is not useful if there are many distinct values.
Summarize with a <i>graph</i> .	Pie chart Bar chart	Pie chart Bar chart	Boxplot Histogram

Revision questions

1. Identify the type of data (nominal, ordinal, interval and ratio) represented by each of the following. Confirm your answers by giving your own examples.

- Blood group
- Temperature (Celsius)
- Ethnic group
- Job satisfaction index (1-5)
- Time taken to go to school from your house
- Number of industrial accidents in a given factory
- Number of cases of each reportable disease reported by a health worker

Next class

Unit III: Introduction to R

NB:

- 1. Everyone MUST come with a PC**
- 2. Make sure you load your phones with enough data for downloads**

